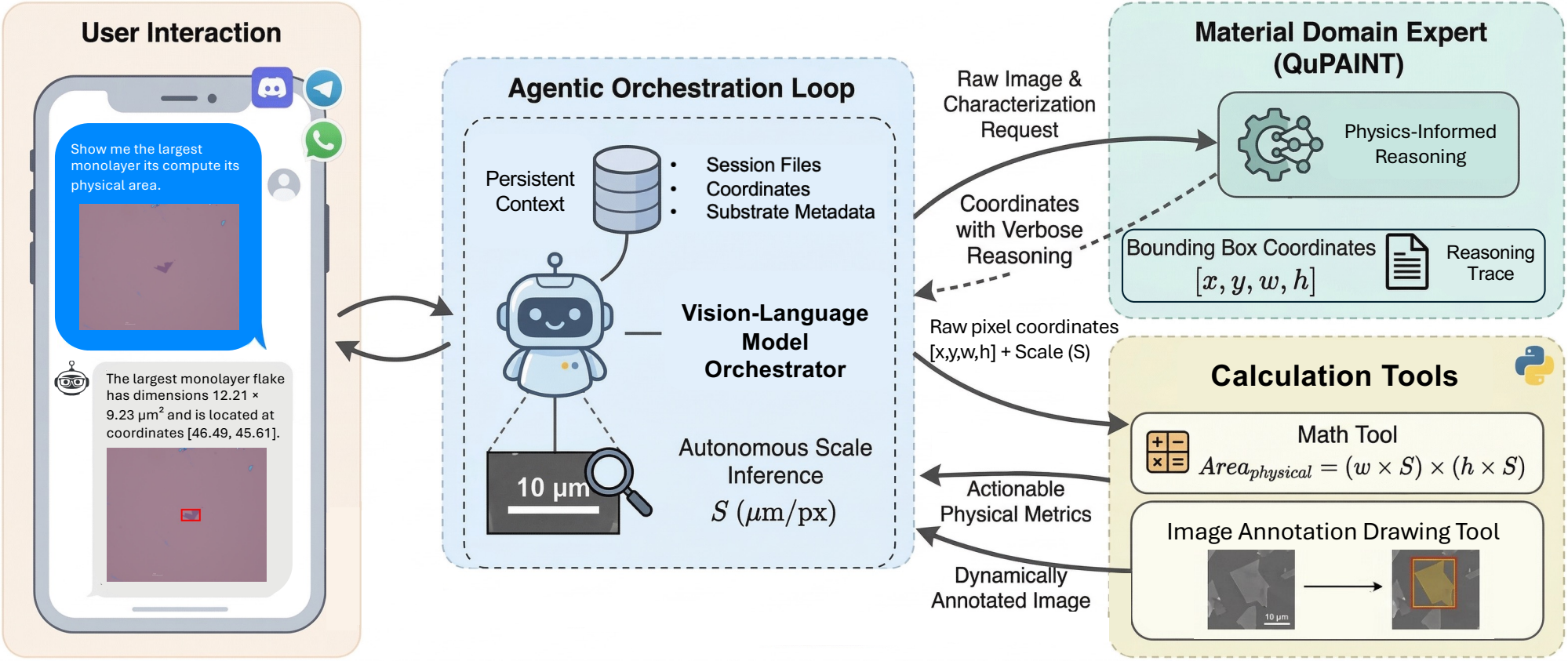
User Interaction
Show me the largest monolayer its compute its physical area.
The largest monolayer flake has dimensions 12.21 × 9.23 μm² and is located at coordinates [46.49, 45.61].
Agentic Orchestration Loop
Persistent Context
Session Files
Coordinates
Substrate Metadata
Vision-Language Model Orchestrator
10 μm
Autonomous Scale Inference
$S\ (\mu m/px)$
Raw Image & Characterization Request
Coordinates with Verbose Reasoning
Raw pixel coordinates [x,y,w,h] + Scale (S)
Actionable Physical Metrics
Dynamically Annotated Image
Material Domain Expert (QuPAINT)
Physics-Informed Reasoning
Bounding Box Coordinates
$[x, y, w, h]$
Reasoning Trace
Calculation Tools
Math Tool
$Area_{physical} = (w \times S) \times (h \times S)$
Image Annotation Drawing Tool
10 μm